

A brief description of the program codes

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This document provides a brief description of each MATLAB file (.m file).

C1_verify.m, C2_verify.m, and C3_verify.m

These program codes are used to verify Theorem 6.1 and Theorem 6.2 for $C_1(T)$, $C_2(T)$, and $C_3(T)$.

C4_verify.m

These program code is used to verify Theorem 6.1 for $C_4(T)$.

Program codes C1_verify.m to C4_verify.m include many subprograms; the relationship among the inclusions is shown in Table 1.

Subprograms	C1_verify.m	C2_verify.m	C3_verify.m	C4_verify.m
create_vertex_list.m	–	–	○	○
create_edge_list.m	○	○	○	○
create_face_list.m	○	○	–	–
set_triangle_alpha.m	○	○	–	–
set_triangle_beta.m	–	–	○	○
F_alpha_0.m	○	○	–	–
F_alpha_1.m	○	○	–	–
F_beta_0.m	–	–	○	–
F_beta_1.m	–	–	–	○
F_beta_2.m	–	–	○	○
create_coefficient_matrix.m	○	○	○	○
L1ab.m	○	–	–	–
L2ab.m	–	○	–	○
L3ab.m	–	–	○	–
L4ab.m	–	–	–	○

TABLE 1 *Relationship among inclusions*

create_vertex_list.m

This program code includes the function `create_vertex_list(n, start_index)`, which maps oblique coordinates of the nodal points to the edge number. For $n=2$, `start_index=1`, the mapping is given as shown in Fig. 1.

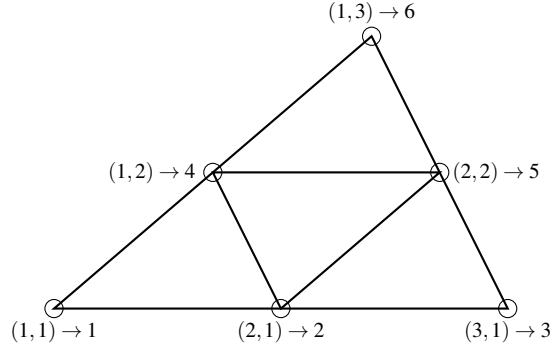


FIG. 1. Mapping obtained using `create_vertex_list`.

create_edge_list.m

This program code includes the function `create_edge_list(n, start_index)`, which maps oblique coordinates of the edge's midpoint to the edge number. For $n=2$, `start_index=1`, the mapping is given as shown in Fig. 2.

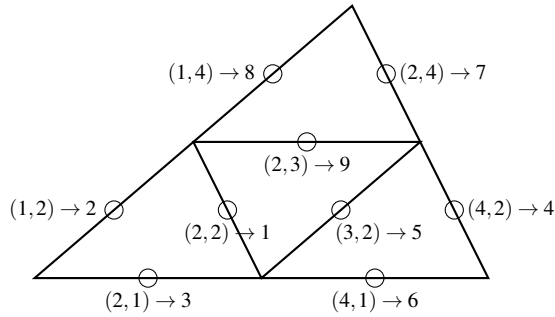


FIG. 2. Mapping obtained using `create_edge_list`.

create_face_list.m

This program code includes the function `create_face_list(n, start_index)`, which maps oblique coordinates of the face's center of gravity to the face number. For $n=2$, `start_index=1`, the mapping is given as shown in Fig. 3.

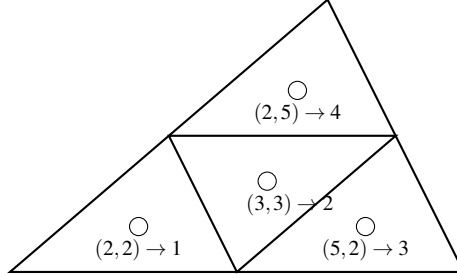


FIG. 3. Mapping obtained using `create_face_list`.

set_triangle_alpha.m

This program code includes the function `set_triangle_alpha`, which returns the mapping from each triangular element to the edge and face numbers that constitute the element.

set_triangle_beta.m

This program code includes the function `set_triangle_beta`, which returns, for each triangular element, the correspondence between the element and its node and edge numbers, together with its orientation.

F_alpha_0.m and F_alpha_1.m

These program codes include the functions `F_alpha_0` and `F_alpha_1`, respectively, which compute $F_0^{(\alpha)}(s, w_1, w_2, w_3, u_0)$ and $F_1^{(\alpha)}(s, w_1, w_2, w_3, u_0)$, as defined in Lemma 5.1.

F_beta_0.m, F_beta_1.m, and F_beta_2.m

These program codes include the functions `F_beta_0`, `F_beta_1`, and `F_beta_2`, respectively, which compute $F_0^{(\beta)}(s, u_1, u_2, u_3, w_1, w_2, w_3)$, $F_1^{(\beta)}(s, u_1, u_2, u_3, w_1, w_2, w_3)$, and $F_2^{(\beta)}(s, u_1, u_2, u_3, w_1, w_2, w_3)$, as defined in Lemma 5.2.

create_coefficient_matrix.m

This program code includes the function `create_coefficient_matrix`, which computes the representation matrix from the corresponding polynomial of quadratic form.

L1ab.m, L2ab.m, L3ab.m, and L4ab.m

These program codes include the functions $L1ab(a, b)$, $L2ab(a, b)$, $L3ab(a, b)$, and $L4ab(a, b)$, which compute $L_j(a, b)$, $j = 1, 2, 3, 4$ defined by equation (3.1).

Lemma5_1.m, Lemma5_2.m, LemmaA_1.m, LemmaA_2.m, LemmaA_3.m, LemmaA_4.m, LemmaA_5.m, LemmaA_6.m, LemmaA_7.m, LemmaA_8.m, LemmaA_9.m, and LemmaA_11.m

Functions of the form `Lemma*_*` in `Lemma*_* .m` are program codes to check the correctness of the corresponding lemmas `*.*`. For example, `LemmaA_11 .m` is the program code used to check Lemma A.11.

In these codes, operations such as $2/3$ are written as `2/sym(3)` to avoid being computed as floating-point numbers. In addition, large integers that cannot be expressed in the range of double-precision floating-point numbers are not accurately converted to symbolic integers if written as is, so to be safe, integers with 10 or more digits are defined using strings such as `sym('1234567890')`.